



Anatomically **Prior-Enhanced X-Ray Network** for Thoracic Disease Classification and Localisation

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Three Critical Gaps in Existing AI Systems

1

**Rare Disease
Blindness**

2

No Anatomical Focus

3

No Visual Explanation

APEX-Net: One System, Three Answers

Balanced Learning

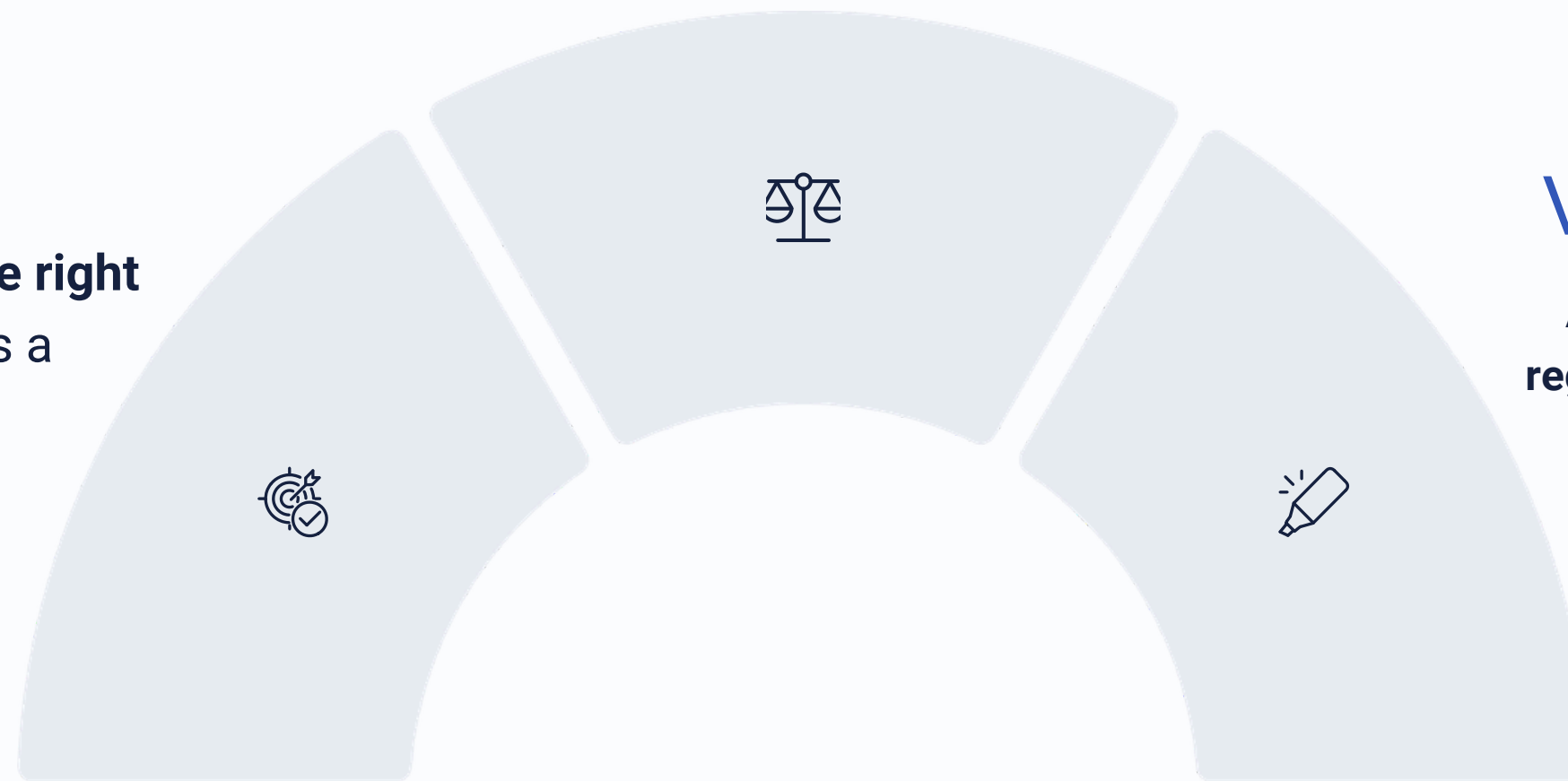
A specialised training strategy that gives **extra emphasis to rare diseases**, reducing missed diagnoses.

Smart Attention

Guides the model to focus on **the right anatomical regions** — just as a radiologist would.

Visual Explanation

Automatically **highlights the affected region** in the X-ray so clinicians can verify every prediction.



QFD

Normal Features

- **Chest X-ray Upload and Analysis**
- **Multi-label Disease Confidence**
- **Heatmap Visualization**
- **Scan History Management**

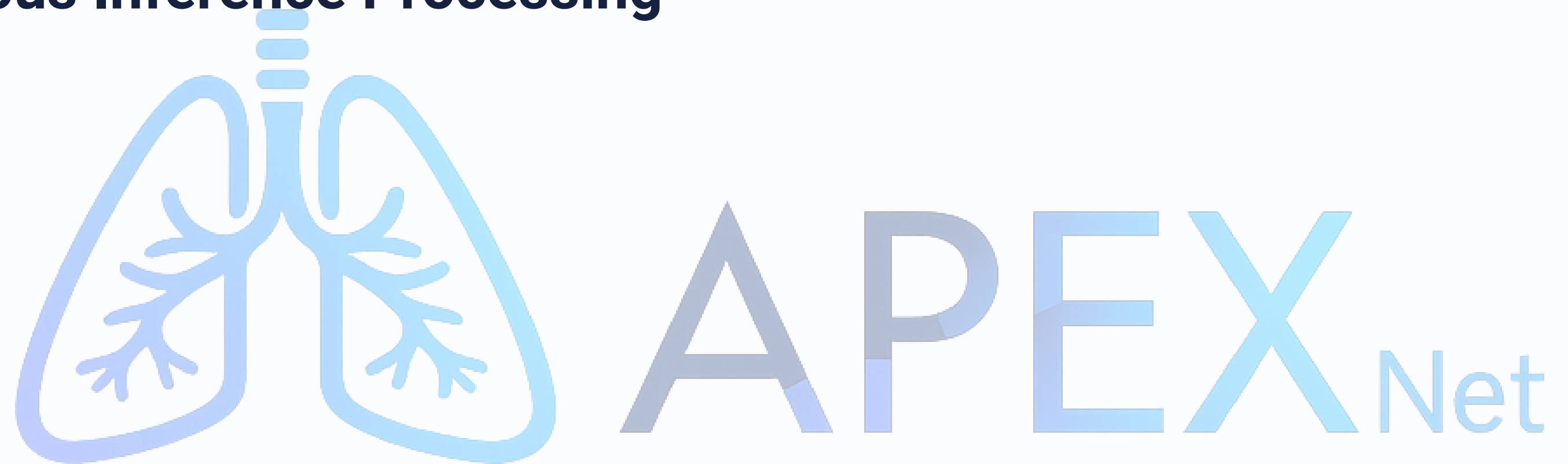


APEX_{Net}

QFD

Expected Features

- **Registration and Authentication**
- **Image Preprocessing and Validation**
- **Asynchronous Inference Processing**



QFD

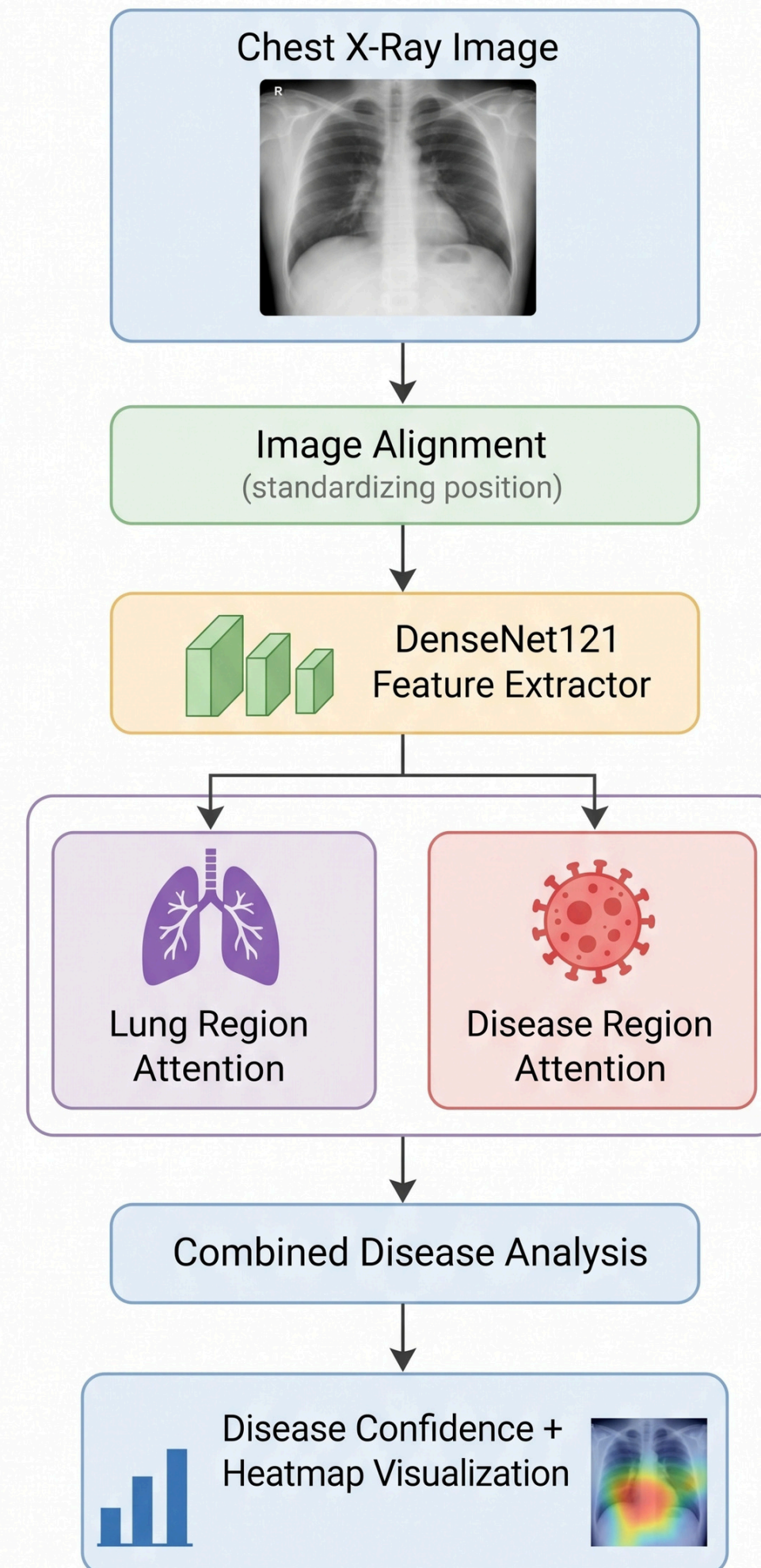
Exciting Features

- **Anatomically-Aware AI Model (APAM Integration)**
- **Rare Disease Sensitivity Optimization**
- **User-Friendly Diagnostic Interface**



METHODOLOGY

How APEX-Net Works



What This Project Delivers



Disease Detection

Accurately classify **14 distinct chest diseases** from a single X-ray image.



Localisation

Highlight **exactly where** in the X-ray each detected disease is located.



Web Application

A working **demonstration interface** that clinicians or reviewers can interact with directly.

Demo

Live project demonstartion



Demo

Video demonstration

The screenshot shows a web browser window displaying the APEX-NET AI Preview application. The browser's address bar shows 'localhost:3000'. The application's header includes the APEX-NET logo, a 'Sign In' button, and a 'Get Started' button. A notification in the top right corner states 'Recording has started'. The main content area features a large heading 'Review chest X-ray model outputs in one workspace.' followed by a subheading 'Upload an image to review generated finding scores and attention heatmaps, then return to saved analyses in your history. For research and educational use only; not for diagnosis or treatment.' Below this text are two buttons: 'Open workspace' and 'Create account'. To the right, a preview of the application's output is shown, featuring a chest X-ray image with a heatmap overlay. A table of findings is displayed next to the image:

Finding	Score
Pneumonia	92.8%
Atelectasis	56.3%
Pleural Effusion	54.2%
Cardiomegaly	12.8%
Edema	8.7%

Below the main content, a 'Workflow' section titled 'From image upload to saved results' is shown. It consists of three steps:

- 01 Upload an image**
Add a chest X-ray image in the analysis workspace.
- 02 Review model outputs**
See generated finding scores and attention heatmaps for the
- 03 Keep a saved record**
Return to completed analyses from your history when

The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 8:12 PM on 12/18/2023.

Challenges

Technical Challenges Faced



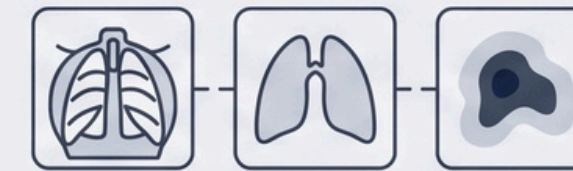
Class Imbalance

Severe class imbalance across 14 disease labels



Uncertain Labels

Uncertain labels in the NIH ChestX-ray14 dataset



Complex Alignment

Accurate alignment of X-rays, lung masks, and disease-prior maps



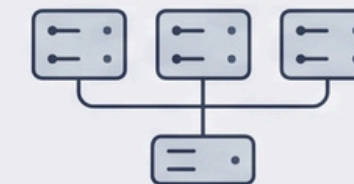
Resource Intensity

High GPU memory and storage requirements for large-scale training



Reproducibility Bar

Reproducing results without access to CheXNet's expert-labeled test set



Server Limitations

Deploying multiple deep-learning models within limited server resources

Thank You

